

[Chathurya V R 251801045-2025-AIDS](#) ▾**C2****Started on** Thursday, 27 August 2026, 8:28 AM**State** Finished**Completed on** Thursday, 17 September 2026, 9:41 AM**Time taken** 21 days 1 hour**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

**Question 1** | Correct | Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than  $\lfloor n / 2 \rfloor$  times. You may assume that the majority element always exists in the array.

**Example 1:**

Input: `nums = [3,2,3]`

Output: 3

**Example 2:**

Input: `nums = [2,2,1,1,1,2,2]`

Output: 2

**Constraints:**

- `n == nums.length`
- `1 <= n <= 5 * 104`
- `-231 <= nums[i] <= 231 - 1`

**For example:**

| Input              | Result |
|--------------------|--------|
| 3<br>3 2 3         | 3      |
| 7<br>2 2 1 1 1 2 2 | 2      |

**Answer:** (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int majorityElement(int nums[], int left, int right) {
4      // Base case: only one element
5      if (left == right) {
6          return nums[left];
7      }
8
9      // Divide
10     int mid = (left + right) / 2;
11
12     // Conquer
13     int leftMajority = majorityElement(nums, left, mid);
14     int rightMajority = majorityElement(nums, mid + 1, right);
15
16     // If both halves have the same majority
17     if (leftMajority == rightMajority) {
18         return leftMajority;
19     }
20
21     // Count both candidates in current range
22     int leftCount = 0;
23     int rightCount = 0;
24
25     for (int i = left; i <= right; i++) {
26         if (nums[i] == leftMajority)
27             leftCount++;
28
29         if (nums[i] == rightMajority)
30             rightCount++;
31     }
32
33     // Combine
34     if (leftCount > rightCount)
35         return leftMajority;
36     else
37         return rightMajority;

```

```
37         return ignoreMajority;  
38     }  
39  
40     int main() {  
41         int n;  
42         scanf("%d", &n);  
43  
44         int nums[n];  
45  
46         for (int i = 0; i < n; i++) {  
47             scanf("%d", &nums[i]);  
48         }  
49  
50         int result = majorityElement(nums, 0, n - 1);  
51  
52         printf("%d\n", result);  
53     }
```

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 3     | 3        | 3   | ✓ |
|   | 3 2 3 |          |     |   |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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